

Building and Evaluating a Retrieval-Augmented Generation System on FinanceBench

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1 Introduction

Retrieval-Augmented Generation (RAG) grounds a language model’s answers in a closed document collection: passages relevant to the question are retrieved from an indexed corpus and the model is instructed to answer from them alone. This report presents the design, implementation, and experimental evaluation of a complete RAG pipeline for financial question answering over SEC filings, evaluated on the FinanceBench benchmark.

Our goal is not a single headline score but a controlled study of which design decisions matter. We sweep 41 retrieval configurations across six axes (chunk size, chunk overlap, document structure preservation, embedding model, number of retrieved passages, and retrieval strategy, including a metadata prefilter derived from the question text), evaluate five free, locally executable generation models, and run oracle-context and closed-book ablations that attribute end-to-end errors to retrieval or to generation. Answers are graded by an LLM judge whose reliability is itself measured against an independent stronger judge.

Three findings stand out. First, restricting the search space with metadata parsed from the question (company and fiscal period) is the single largest retrieval improvement we measured, worth 13 to 16 points of recall at every k, far more than any chunking or embedding change. Second, a general-purpose cross-encoder reranker and BM25, both standard components elsewhere, actively hurt on this corpus of dense financial tables. Third, the oracle ablation shows that .

2 Corpus Description

FinanceBench (Islam et al., 2023) is an industry benchmark for question answering over public financial filings. Its open source split contains 150 questions over 84 documents: 10-K and 10-Q filings, 8-K reports, and earnings releases from 40 public companies, fiscal years 2015 to 2023. Every question carries a gold answer and one to three evidence items, each annotated with the exact supporting text and its zero-based page number. Questions come in three types of 50 each: metrics-generated (read a value off a statement), domain-relevant (analyst questions requiring judgement), and novel-generated (unusual questions unlikely to be memorised), spanning information extraction, numerical reasoning, and logical reasoning.

The benchmark ships a document catalog with source URLs rather than the documents them-

selves, and about a quarter of those URLs have rotted since publication. Our downloader retries with SEC-compliant request headers and falls back to the official FinanceBench GitHub mirror; 61 documents arrived from their original URLs and 23 from the mirror, for full coverage of all 84 documents and therefore all 150 questions. A committed manifest records the provenance, size, and SHA-256 of every file.

Text is extracted per page with PyMuPDF: 12,013 pages and 40.0M characters across the 84 documents. Cleaning is deliberately conservative (unicode NFKC normalisation, control-character removal, whitespace bounding) because aggressive normalisation destroys the row and column adjacency that financial tables depend on. Only 78 pages (0.65 percent) yielded fewer than 20 characters, indicating scanned or graphical pages; no document required OCR. Pages are the atomic unit of ground truth throughout: retrieval is scored against the annotated evidence pages, and a text-level containment measure catches evidence lost inside a retrieved page.

3 Methodology

3.1 Segmentation

Three chunkers cover the structure-preservation axis. The token chunker slides fixed windows over the whole document, ignoring layout. The page chunker treats the PDF page as the atomic unit and splits only oversized pages. The structural chunker packs whole paragraphs greedily up to the token budget, never cutting through one. All chunk boundaries are measured with one fixed reference tokenizer (bert-base-uncased) regardless of the embedding model, so that when we compare embedding models they see literally identical chunks. Every chunk carries its document name and the page span it covers.

3.2 Retrieval

Chunks are embedded with sentence-transformer models, using the query and passage prefixes each model’s authors document (E5 and BGE are asymmetric; omitting their prefixes measurably degrades recall and would contaminate the comparison). Normalised embeddings are indexed in FAISS and searched by inner product. Three alternative strategies are implemented behind the same interface: Okapi BM25; a hybrid that min-max normalises dense and BM25 scores and combines them with weight alpha; and cross-encoder reranking (ms-marco-MiniLM-L-6-v2) applied to a dense candidate pool of configurable size.

The metadata filter restricts candidates to documents whose company is named in the question, narrowed further by fiscal year when one is mentioned. It is built by matching the catalog’s 40 company names (plus a small alias list) against the question string after both are lowercased and stripped of punctuation; 127 of the 150 questions name their company verbatim and the filter deactivates itself on the rest. It reads nothing from the gold record except the question text, so it uses only information a deployed system would have.

3.3 Generation

Answers are produced by free, locally executable instruction models served by Ollama, prompted to answer strictly from the numbered passages, to cite every fact as [n], to state numeric answers with units, and to abstain with a fixed phrase when the passages are insufficient. The context

window (16,384 tokens) sits well above the largest measured prompt, and every response’s token accounting is checked against the window so silent truncation cannot pass unnoticed; zero truncations were flagged across all runs. Three context modes support the ablations: retrieved (the RAG system proper), oracle (the gold evidence passed directly, removing retrieval from the equation), and none (closed book, bounding how much of the score could be pre-training memorisation).

3.4 Evaluation

Retrieval is scored at page level against the annotated evidence:

- recall@k: some retrieved chunk covers a gold evidence page;
- strict recall@k: a single chunk contains at least half of an evidence item’s 5-grams, i.e. the evidence text itself was retrieved, not merely its page;
- document accuracy@k, MRR@k, and nDCG@k with page-level relevance;
- evidence coverage@k: the mean fraction of evidence 5-grams present anywhere in the retrieved set, the measure that exposes chunk-boundary evidence loss.

Generation is scored deterministically with scale-tolerant numeric accuracy (FinanceBench answers are mostly dollar amounts, percentages, and ratios; the scale-tolerant variant forgives unit mismatches such as millions versus raw dollars but not wrong digits), SQuAD-style token F1, citation rate and citation validity (whether a cited passage actually covers a gold page), and abstention rate. An LLM judge (qwen2.5-14B, temperature 0, JSON-constrained output) grades correctness (0 to 2) against the reference answer and groundedness (0 to 2) against the passages shown to the model.

Because an LLM judge is an assertion until measured, we validate it by re-judging a seeded random sample of answers with an independent, stronger judge (gpt-5.5 via the Codex CLI) and reporting exact agreement and Cohen’s kappa.

4 Experimental Setup

All heavy computation ran on a RunPod A100-SXM4-80GB pod (CUDA 12, PyTorch 2.9, Python 3.12); development, unit tests (64 tests), and the judge cross-check ran on a laptop. The retrieval grid comprises 41 configurations: a 4x4 chunk size (128, 256, 512, 1024 tokens) by overlap (0, 10, 25, 50 percent) sweep, three structure modes, six embedding models (all-MiniLM-L6-v2, gte-base, e5-base-v2, bge-base-en-v1.5, e5-large-v2, bge-large-en-v1.5), k in {1, 3, 5, 10, 20}, BM25 at two k values, hybrid at alpha in {0.3, 0.5, 0.7}, reranking with pools of 20, 50, and 100, and the metadata filter at three k values plus one filter-plus-rerank combination. Sweeps are one-factor-at-a-time around a fixed baseline (token chunks of 512 with 64 overlap, bge-base, dense, k=5) so each comparison isolates one decision. The full grid embeds roughly 460k chunk instances and completed in about 100 minutes on the A100; every run is cached under a hash of its configuration and is resumable.

Generation used the best retrieval run eligible for the prompt budget (the metadata-filtered dense run, whose top 10 passages reach 0.627 recall) across five models spanning the four families named in the brief and two sizes of one family: llama3.1-8B, mistral-7B, qwen2.5-7B, phi4-14B, and qwen2.5-14B (all Q4 quantised). Ablations add oracle and closed-book runs for the two

qwen sizes and a k=20 context-size variant for the strongest model. Generation is deterministic (temperature 0) and every run checkpoints every ten questions.

5 Results

5.1 Retrieval

Table 1: headline retrieval results (full 41-row table in results/tables/retrieval_results.md).

Configuration	k	Recall@k	Strict@k	MRR@k	ms/query
dense + filter	20	0.707	0.507	0.307	57
dense + filter	10	0.627	0.407	0.301	37
dense, plain	20	0.587	0.427	0.204	13
dense + filter	5	0.493	0.300	0.284	25
dense, plain	10	0.500	0.333	0.198	8
e5-large-v2 (best embedding)	5	0.453	0.327	0.316	7
rerank, pool 20	10	0.447	0.340	0.203	178
baseline (bge-base)	5	0.333	0.240	0.177	6
hybrid, best alpha 0.7	10	0.320	0.233	0.152	341
BM25	10	0.153	0.127	0.096	229

Metadata filtering dominates every other intervention. At each k the filter adds 13 to 16 recall points over plain dense search (0.333 to 0.493 at k=5, 0.500 to 0.627 at k=10, 0.587 to 0.707 at k=20) and roughly halves the rank of the first relevant hit (MRR 0.30 versus 0.20). The mechanism is visible in the error taxonomy: wrong-document retrievals drop to 2 percent of questions. Filtering costs a few milliseconds and no training; on a corpus whose documents are near-duplicates of each other in vocabulary (every 10-K contains an income statement), identity metadata carries information that embeddings alone recover only partially.

Chunking (Figure @fig:chunking). The chunk sweep spans 0.293 to 0.373 recall at k=5, a smaller range than the filter’s effect at any k. A 10 percent overlap is best at every size (mean +0.04 over no overlap); 256 and 512-token chunks lead on recall. Evidence coverage tells the other half of the story: it doubles from 128-token chunks (0.11 to 0.15) to 1024-token chunks (0.26), because small windows retrieve the right page but hand the generator a fragment of the evidence. This tension (small chunks rank better, large chunks carry more of the evidence once found) motivates the 512/64 compromise used downstream.

Structure preservation. Page-bounded chunks beat token windows (0.367 versus 0.333) with structural packing between (0.347). The gold labels are page-level, and financial statements are laid out one per page, so respecting the page boundary aligns chunks with both the annotation unit and the document’s own semantics.

Embedding models. e5-large-v2 is the strongest (0.453 recall, 0.316 MRR, the best MRR of any configuration including filtered ones), followed by bge-large (0.413), e5-base (0.387), bge-base (0.333), gte-base (0.313), and MiniLM (0.253). The E5 family outperforms at equal parameter

count, and scale helps within each family. Notably, swapping bge-base for e5-large is worth +0.12 recall, slightly less than what the metadata filter adds on top of any of them.

k. Recall grows monotonically from 0.093 at k=1 to 0.587 at k=20 with no plateau, so the retriever keeps finding gold pages deeper in the ranking; the cost is prompt length and distraction for the generator, which the k=20 generation ablation quantifies below.

Negative results: BM25, hybrid, and reranking. BM25 reaches only 0.153 at k=10, one third of dense retrieval, and it drags every hybrid mixture below pure dense (best hybrid 0.320 at alpha 0.7). Filings are boilerplate: the terms in a question (“capital expenditure”, “fiscal 2018”) appear on dozens of pages of every document, so term frequency barely discriminates, while paraphrase (the statement line reads “purchases of property, plant and equipment”) defeats exact matching outright. The ms-marco cross-encoder also underperforms the dense ranking it reranks (0.380 to 0.447 versus 0.500 at k=10), and enlarging its pool makes it worse, the opposite of the usual pattern: trained on web passages, it mis-scores dense financial tables, and a larger pool only gives it more distractors to promote. Both components would be net-negative additions to this system despite their standard-recipe status.

5.2 Generation

6 Error Analysis

6.1 Retrieval failures

At the filtered k=10 operating point, 56 of 150 questions (37.3 percent) miss the gold page. The taxonomy separates three mechanisms:

Failure mode	Count	Share
Wrong document retrieved	3	2.0%
Right document, wrong page	53	35.3%
Gold page hit but evidence text mostly absent	19	12.7%

Discrimination between documents is essentially solved by the filter; what remains is ranking inside a single filing, where the query must distinguish the income statement from ten pages that mention the same terms. The third row is invisible to page-level metrics: the right page was retrieved but the chunk boundary cut the evidence away (median coverage 0.03 among those 19), which is exactly the failure the evidence-coverage metric was added to catch.

Failure rates differ sharply by question type: 16 percent for metrics-generated questions (whose phrasing mirrors statement line items) versus 56 percent for domain-relevant and 40 percent for novel-generated questions, whose analyst phrasing shares little vocabulary with the evidence.

7 Discussion

8 Conclusion